



Jiyeon Joung

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<https://github.com/sanananan3>

EDUCATION

Hanyang University ERICA Campus (2020 - 2023)

Department of Mechanical Engineering

SungKyunKwan University (2023 - 2025)

Computer Science and Engineering

• Total GPA of 3.92 / 4.5, Major GPA of 3.88 / 4.5

SKILLS

- Python
- C++
- HTML, CSS
- React, R3F
- ROS
- Issac
- MuJoCo
- Gazebo
- Pytorch

RESEARCH INTERESTS

Efficient ML for Edge device

For real-time inference & low-power computing, I Focus on model compression and optimization using quantization methods

DL Applications & Developments in robotics

Learning-based visuomotor control across Imitation Learning, Reinforcement Learning (End-to-end pipelines, sim2real transfer)

WORK EXPERIENCE

2026.1 - 2026.3

Huge (Humanoid generalization) Lab, KAIST
Intern Researcher

- Support the development of a new neural simulator
- Propose evaluation metrics adopted for benchmarking

2024.12 - 2026.2

Korea Institute of Science and Technology
Intern Researcher

- Reduce Inference latency via DL architecture improvements and optimization
- Implement Diffusion Policy on Simulator and Real Robot
- Implement Action Chunking with Transformers on Simulator and Real Robot
- Reinforcement Learning for mobile manipulator
 - PPO, Hierarchical RL

2024.8 - 2024.12

IRIS (Intelligent & Resource-efficient Image processing & Systems design) Lab, Sungkyunkwan University
Undergraduate Researcher

- Develop multi-bit PTQ of weights and activations for a range of CNN models

PUBLICATION

AgileDP: Quantizing Diffusion Policy via Dynamic Scaling and Selective Distillation

Jiyeon Joung, Namyoon Kim, Seungseop Lee, and Keunwoo Jang
(Under review.)

Scalable Neural Simulator for Articulated Rigid Bodies

Jan. 2026 ~ Mar. 2026

- Research assistant at HuGe Lab, KAIST
- Conducted technical ablations for a novel neural simulator and built an adopted evaluation pipeline for benchmarking against Isaac, RaiSim, MJX, MuJoCo-CPU, and ours.

Diffusion Policy for Robotics Manipulation on Franka Panda

April. 2025 ~ Feb. 2026

- Research projects conducted at KIST (Korea Institute of Science and Technology)
- Implemented Diffusion Policy on a real Franka Emika Panda arm with a overall workflow.
- Evaluated multiple inference strategies (e.g., door opening, pick-and-place) and built our own MuJoCo-based sim-to-real workspace.
- Inference optimization for Diffusion Policy with real-world deployment on G1 and Franka Panda, published as a paper.

Action Chunking with Transformer on Franka Panda

Feb. 2025 ~ April. 2025

- Research projects conducted at KIST (Korea Institute of Science and Technology)
- Adapted ACT for real-robot use beyond ALOHA robotics platform, creating a reusable workspace.
- Used polynomial interpolation instead of temporal ensemble to lower GPU overhead and enable efficient inference with smoother robot motion.

Crop Management System using Generative AI and Computer Vision

Mar. 2025 ~ June. 2025

- 2025-Spring Semester Graduation Project, Department of Computer Science and Engineering, SKKU
- Built an application to support novice farmers with crop management recommendations.
- Developed a CVAE-based model to recommend optimal greenhouse conditions (temperature, humidity, airflow).
- Integrated YOLOv8 for pest/disease detection and classification.

Hierarchical Reinforcement Learning (HRL4IN) for Mobile Manipulation

Dec. 2024 ~ Feb. 2025

- Research projects conducted at KIST (Korea Institute of Science and Technology)
- Designed hierarchical policies (task generator + control) to overcome PPO's limitations in multi-phase tasks.
- Demonstrated sub-task success (e.g., grasping door handle) in MuJoCo simulations with Franka Panda.

PTMQ-Plus : Multi-bit Weight & Activation Post-Training Quantization

Aug. 2024 ~ Dec. 2024

- Research projects conducted at IRIS Lab, Department of ECE, Sungkyunkwan University as undergraduate students
- Developed PTMQ-Plus, extending multi-bit quantization to both weights and activations.
- Enabled efficient CNN and ViT inference for edge AI, reducing memory and power demands.

Computer Vision-based Road Damage Detection Service "Saveway"

June. 2024 ~ Aug. 2024

- Team projects at K-SW Empowerment Bootcamp, Sungkyunkwan University
- Fine-tuned YOLOv9-s for damage detection and utilized ESRGAN for resolution enhancement.
- Built pipeline from mobile app capture → server processing → MobileNet-based detection → web dashboard visualization.

Computer Vision-based Sign Language Learning Website "Slang"

Mar. 2024 ~ June. 2024

- Team projects conducted at Software Capstone Project Course, Sungkyunkwan University (Spring 2024)
- Built a web platform using Google MediaPipe for precise sign gesture recognition and learning.
- Developed interactive 3D UI with Three.js and React Three Fiber as Front-end team leader.

Sensor-based Autonomous Driving Software

Mar. 2024 ~ June. 2024

- Team projects conducted at Smart Car Integrated Design Course, Sungkyunkwan University (Spring 2024)
- Programmed ESP32 with C++ (Arduino IDE) to control IR and ultrasonic sensors and design driving logic.
- Implemented real-time path planning for obstacle avoidance, T/parallel parking, and intersection crossing.
- Developed a Kotlin-based service app for disabled passengers, awarded in final presentation.

Deep Learning-based ROS2 Autonomous Driving Project

Jan. 2024 ~ Feb. 2024

- Team projects conducted at Future Automotive Microdegree Program, Sungkyunkwan University
- Collected and annotated lane images using Roboflow for segmentation and YOLOv8, and later extended YOLOv11.
- Implemented lane detection on a 1/10-scale car with Arduino-based control for high-speed driving.
- Focused on SW-driven pipeline (ROS2 + deep learning) to achieve robust autonomous navigation under limited HW.

Entertainment Betting Webpage "Pick The Gold"

Dec. 2023 ~ Jan. 2024

- Team projects conducted at KAIST "몰입캠프"
- Designed full-stack web service for weekly prediction betting (React & TypeScript frontend, Node.js & MySQL backend).

AWARDS AND HONORS

- 2025.08 **Encouragement Award**, The 3rd Sungkyunkwan University Autonomous Driving Software Competition, 2025
- 2025. 07 **3rd place**, Research Paper & Project Excellence, Department of Software, Sungkyunkwan University 2025
- 2025.02 **Academic Scholarship**, Future Automotive Microdegree, SKKU (Winter 2024)
- 2024.09 **2nd place**, Certificate Of OpenSource SW Activity
- 2024.08 **3rd place**, K-SoftVation Showcase
- 2024.08 **Academic Scholarship**, Future Automotive Microdegree, SKKU (Spring 2024)
- 2024.07 **Challenge Award**, The 2nd Sungkyunkwan University Autonomous Driving Software Competition, 2024
- 2024.02 **3rd place**, 2023 Winter ROS2-based Autonomous Driving hackathon
- 2021.11 **3rd place**, 2021 ERICA IoT Online English Speech Competition

OTHERS

CERTIFICATIONS

- 2024.09 **Engineer Information Processing**
- 2024.06 **SQLD : SQL Developer**
- 2023.11 **ADsP : Advanced Data Analytics Semi-Professional**
- 2021.09 **Korean History Proficiency Test, Level 2**
- 2021.09 **Computer User in the field of spreadsheet and database, Level-1**

QUALIFICATIONS

- 2025.09 **TEPS** (337)
- 2024.02 **TOEIC** (865)